

Firearms Background Checks

Introduction

This report will look into the statistics of firearm background checks using the National Instant Criminal Background Check System (NICS) in the United States between 1998 and 2024. It covers all 50 states and the District of Columbia, as well as to different kinds of transactions, including handgun, long-gun and permit checks. After cleaning and preparing the data, statistical and time-series analysis of the data were conducted to examine the trends, seasonality, state variations, and demand patterns. The discussion presents a plausible view of the operation of the background check of guns over time.

Data Overview and Preparation

This report is an examination of firearm background check statistics at the National Instant Criminal Background Check System (NICS) in the United States in 1998-2024. The data covers 50 states, District of Columbia, and 24 various types of transactions, including handgun checks, long-gun checks, permits, and multiple-guns transaction. Missing data was addressed, date format conversion, categories standardised, territories with irregular reporting were eliminated, and records across time periods harmonised to clean the data. U.S. Census population density data was included to compare per-capita data, and time-series techniques and trend analysis were used to analyse time-series behavior, visualised seasonal variations, and demand patterns. The resulting cleaned data is a sound foundation of the five analytical sections that follow.

Methods:

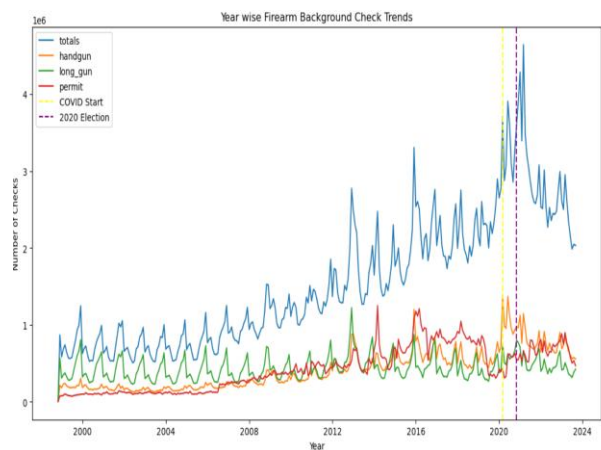
The monthly NICS data between 1998 and 2024 was standardised and aggregated on a national and state level and cleaned. All the long-term trends, seasonality, and event-related peaks were analysed throughout the entire period to determine the baseline. The variation on the state level in the nature and volumes of transactions was examined to determine structural differences in the nature of state reporting and classification of NICS checks. Population density data of the US census was included to allow using them per-capita to compare volumes of background checks by the state. Month-on-month growth rates and in-year regression slopes were calculated between 2019 and 2023 to

analyse the demand variation and composition difference of each type of checks statewide. Lastly, the national monthly series were used in SARIMA and the Random Forest models to produce 12-month forecasts for checks and compare model output with held out test data.

01 Annual Firearm Background Check Trends

This section explores the monthly patterns of firearm background checks in the United States in 1999 to 2024 by the three categories namely: total checks, handgun checks, long-gun checks, and permit related checks. The objective is to study the long-term trends, seasonality and the impact of major socio-political events on the firearm buying behaviour. The growth rate is higher in the period 2012-2019, which is likely explained by policy debate and anticipatory buying behaviour. Its steep incline is observed in early 2020 when the COVID-19 pandemic erupted, which is likely because of panic buying, misunderstandings, and the increased demand in personal protection. The other spike is observable in the years surrounding the 2020 election, which could be due to the uncertainty of politics and the fear of future gun laws.

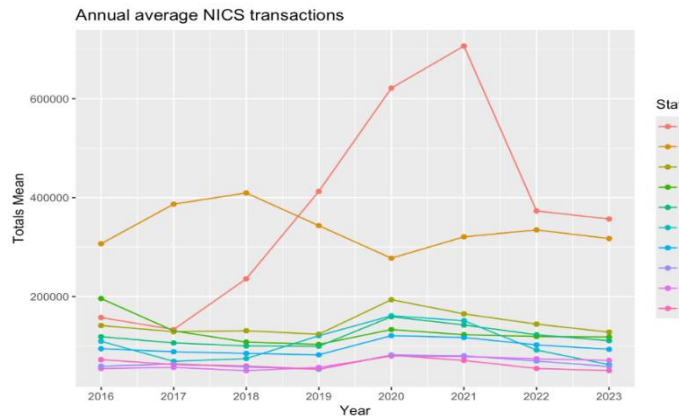
Once these peaks are surpassed, the background checks drop between 2021 and 2024, albeit at larger levels than they were prior to the pandemic, suggesting that the effects continue to impact purchasing behaviour. Its seasonal patterns exist throughout the entire data set and the recurring peaks tend to be found in the last quarter of each year.



02 The Visualising Rate of Change in Total Background Checks Across States and Transaction Types

To visualise the changes in total number of NICS background checks by state, original data was limited to years from 2016 to 2023, which all have full entries for the most prevalent transaction types.

This plot shows annual transaction averages for 10 states with highest mean NICS permit checks between 2016 and 2023. Most states have similar patterns in changes to transaction numbers, staying under 200,000 without major fluctuations. However, Illinois and Kentucky consistently show higher transaction averages, with Illinois reaching totals of over 600,000 in 2020 and 2021. Despite both states averaging higher than the rest, for the most part they do not follow the same pattern in year-to-year changes, unlike the rest. The vastly different average figures for the likes of Illinois and Kentucky versus the rest of the states can be explained by local gun laws and other external factors, which are explored further in the report.



Plot 2 : NICS Transactions by Type and Year

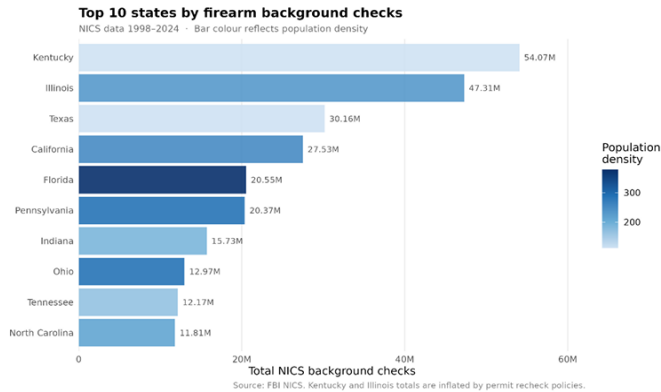
The same reduced dataset was used for this plot, also limiting the columns to Permit, Permit Recheck, Handgun and Long Gun — most common transaction types across the states. Considering the findings of the previous plot, i.e. the similarity in average totals pattern for all states but Illinois and Kentucky, the data was further reduced to top 5 states by average total transactions (Figure Figure 1).

Similarly to the annual average transaction plot, California, Florida and Texas show likeness in year-to-year distributions of transactions, with a complete lack of permit rechecks for all states bar California in 2021-2023. For Illinois and Kentucky however, most of the transactions in certain years are permit rechecks, with very few handgun or long

gun checks, which comprise majority of transactions for other states. This plot reflects differences in classification of background checks performed by NICS across the states, which once again are affected by local gun laws for each state.

03 Population Density and Firearm Background Checks Across U.S. States

NICS background checks are conducted between 1998 and 2023 across states and territories in the United States and give a broad-scale view of the activity of firearm background checks over the long-term nationwide. The bar length is used as a representation of total checks and the bar color of population density in the chart, so both variables can be compared in a single visual without having to plot them separately. The top checks are registered in Kentucky (54 million) and Illinois (47 million), but since they are inflated by regular permit renewal, they are not as reliable in comparison. Texas and California follow, with larger volumes accordingly to their high populations, and the tightening gun laws in California might lower totals. The relationship between Florida and Pennsylvania has a better population-to-checks relationship. States with lower density like Indiana and Tennessee have less number of checks. On the whole, population density affects check volumes, while state permit policies and firearms legislation are strongly influencing.



04 State-Level Firearm Demand: Growth, Trend and Composition

State-level firearm demand was measured by summing check types of handgun, long gun, multiple, permit, and other these categories best represent checks linked to newly entering firearms (Nemerov, 2018). Growth was computed as the month-over-month percentage change of demand (2019-2023). U.S. territories were removed as they differ in reporting and policy. The District of Columbia was retained since it offers a useful

comparison point. The growth, trend and slope are presented in interactive dashboard view available in <https://rajarajeswari-sankar-data-science-2026.shinyapps.io/nics-firearm/>

Average Monthly Growth in Demand (Within Year): This growth rate map shows how firearm demand varies across states each year (Figure Figure 2). The RdBu scale reflects blue for increase in demand, decline in red, and white for little change. It shows clear variation between states in 2019. Mississippi has fast growth due to buying ahead of state elections. In 2020 most states show only mild blue with few states in darker blue because the monthly average smooth out the demand spike of early 2020, so the COVID surge does not appear clearly. After the 2020 surge, many states show less growth in demand. Then differences between states appeared again in 2022. Oregon shows high growth because of Measure 114 (Bollman et al., 2025). By 2023 most states had decreased or no growth in sales. This shows that the market is becoming stable.

Demand Trend Within Year (Slope): The slope map shows whether demand is rising or falling over the year (Figure Figure 2). Looking at both maps helps us to see how the states are changing over time. On the slope map, darker purple shades show negative slopes, yellow shades for positive slopes, and mid green tones for slight change. If we look at the map for 2019 most states are similar while Kentucky shows negative trend. In 2020 the slope map shows rising patterns across the state especially Florida, Michigan and Pennsylvania, these differences were hidden in growth map. Nearly most states show a negative trend in 2021 and near-zero in 2022, whereas Illinois continuously shows positive trend. Then in 2023 most states are in dull green shades show a negative trend. Illinois shifts to the negative slope after having two years of leading in positive trend. Together, they provide pair of information: growth is the scale of the change, and slope is the direction. An example of the high growth with a negative slope would imply there is a temporary spike and then fall and a high growth with a positive slope would imply sustained growth.

Categorical-Composition Bar Chart: The bar chart shows what is behind the firearms demand in each state (Figure Figure 2). It breaks down the proportion of each category that derives the demand. This information helps us understand the growth and trends we see on maps. For example, in 2020 Michigan had high growth, by looking at the chart we can see that handguns checks have high

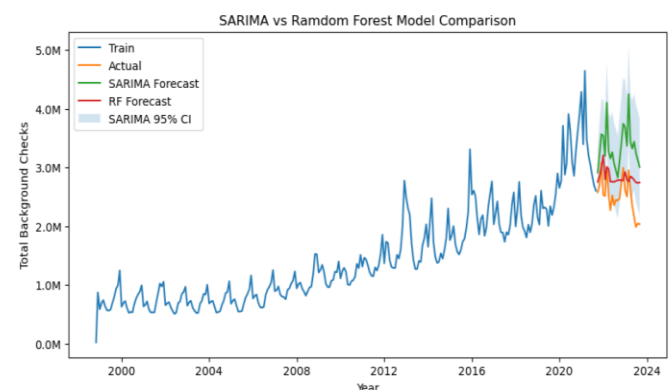
proportions. The permit share in Illinois is relatively large which makes it stand out in 2021.

Firearm demand highly sensitive to major event such as elections, COVID and policy changes (FigureFigure 3). NICS totals include re-permit and other admin checks, they are not accurately representing purchases. Focusing on direct purchase checks shows that handguns is dominating in most states, while high permit shares reflect state procedures rather than real demand.

05 Predicting future background check trends

Plot 1: SARIMA vs Random Forest – Model Comparison

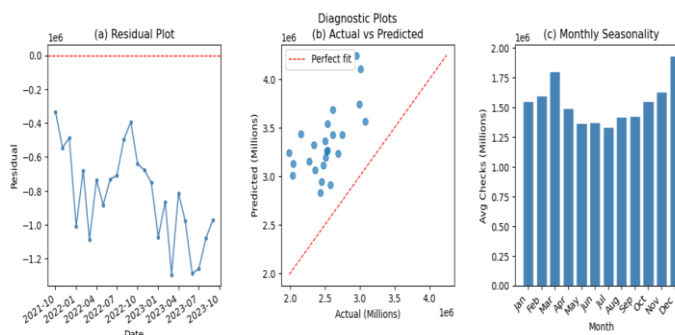
Although the SARIMA model does well capturing the seasonal peaks, it has an average underestimation of the actual monthly number of background checks during the test period by 2.9M-4.1M against the actual demand of 2.0M-3.1M. However, the predictions made by the Random Forest model exhibit less variability and no season peaks at all. This overfitting can be attributed to the fact that there are extreme peaks in the training set that were seen as normal according to the model due to the COVID pandemic. This resulted in further overestimations in the period of forecasting when the demand began to decline. In this case, it can be recommended that SARIMA is the superior model because it is able to better model both seasonality and trends as compared to Random Forest. Seasonal models are more effective when the data is periodic, and the series of monthly checks of NICs are clearly and regularly seasonally behaved, and SARIMA should be able to deal with them.



Plot 2 : Diagnostics Plots

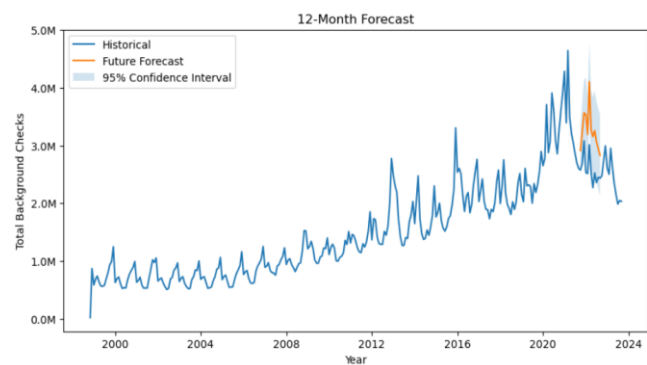
The three diagnostic plots are combined to make sure the SARIMA forecasting model is valid. The plot of residual (a) shows stable residual throughout the time without any drastic changes but all residual values are negative, which means that the model

overestimates the results consistently due to the artificial high baseline of 2020-2021 trainings. Notably, the trend of error is not on the rise implying that model is stable and does not deviate. The majority of the actual values versus predicted values plot (b) points have been plotted along the ideal fit line that shows that the forecasting model can predict general magnitude and trend of demand with an overestimation of bigger values. The seasonality graph (c) distinctly shows that there is a strong annual cycle of changes with the highest number of background checks in December (approximately 1.93 M people) and the lowest number in July (approximately 1.33 M people).



Plot 3 : 12-Month Forecast

The 12-month forecast plot indicates that the background checks on gun ownership will be quite high in the nearest future, in the same seasonal distribution as the previous ones, with minor fluctuations, meaning that the seasonal behaviour is also likely to persist. The general direction seems also to be more stable than the soaring numbers witnessed in and around 2020-2021, indicating that the number of background checks may have stabilized following the initial boost. It is estimated that monthly background checks will range between 2.5 million and 4.1 million with the use of SARIMA model. The highest month is projected to be in March 2022 (4.1M) and the lowest month will be in September 2022 (2.8M). Findings show that there are no post-pandemic changes but with a seasonal rise towards the later part of the year. The forecast however fails to take into consideration any surprise policy changes or socio-political events.



Summary and Conclusion

The report explores the history of gun background checks in the United States during the period 1998 to 2024, its trends, and variability of the states, change in demand and forecasting. Results suggest that the number of checks has risen in the long run, and there have been substantial peaks during the COVID-19 pandemic, during the 2020 election, policy changes and then declined, nevertheless, at a higher rate than prior to the pandemic. The states are very varied. Kentucky and Illinois are difficult to compare since they are outliers owing to the variations in policy-based reporting. The volumes of the background checks are not determined by population only but also by the state laws, demographics and legislation. The market largely stabilised in 2023 and most states saw negative growth in 2021 after the 2020 surge. SARIMA was better than the random forest since, unlike the other forecasting models, it could predict 12 months (Figure Figure 4) with higher accuracy since it was able to capture seasonality and trends. Overall, politics, socioeconomic factors and policy have a heavy impact on firearm background checks.

Reference

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[5] U.S. Census Bureau (2024). Population Estimates Program. Available at: www.census.gov

Figures:

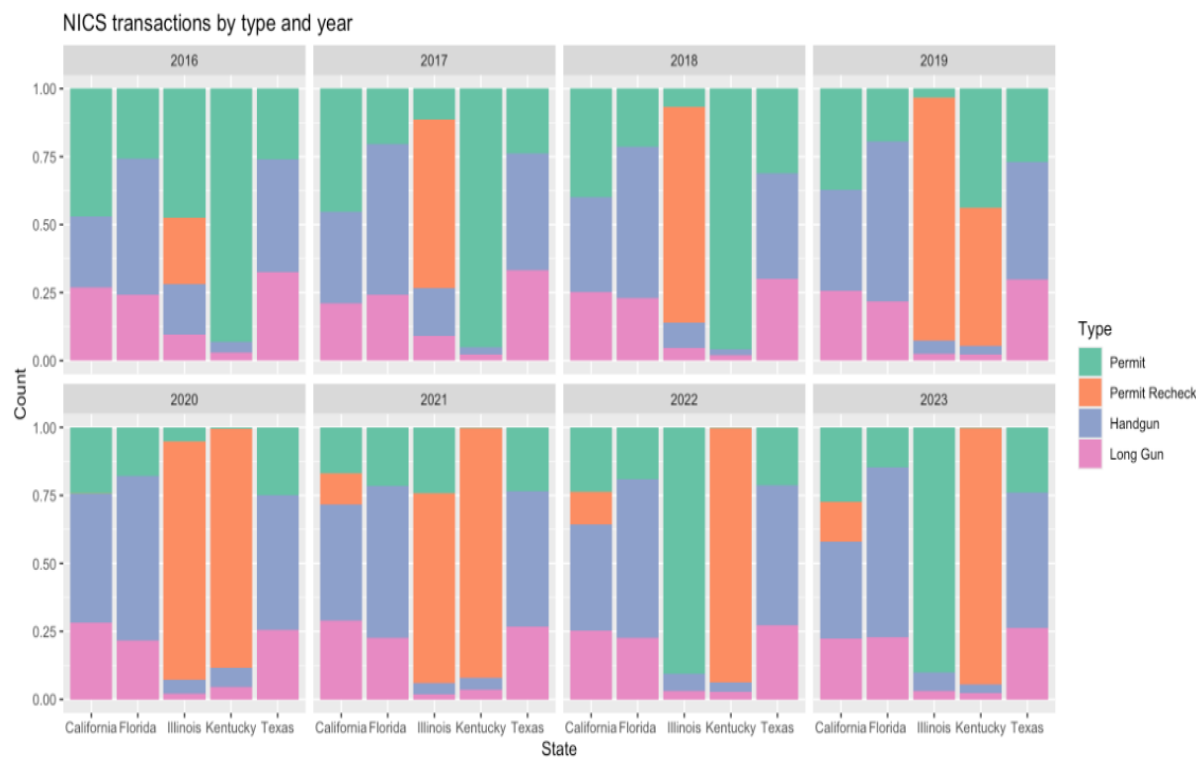


Figure 1. NICS Transaction by Type and Year

State-Level Firearm Demand Analysis: Growth, Trends, and Composition

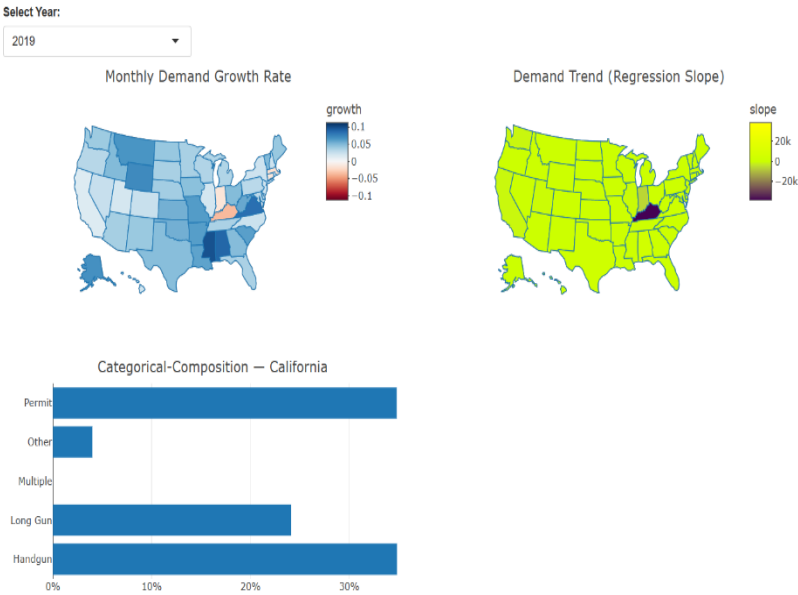


Figure 2. Interactive Dashboard Screenshot

Composition of Demand Checks (2019–2023)

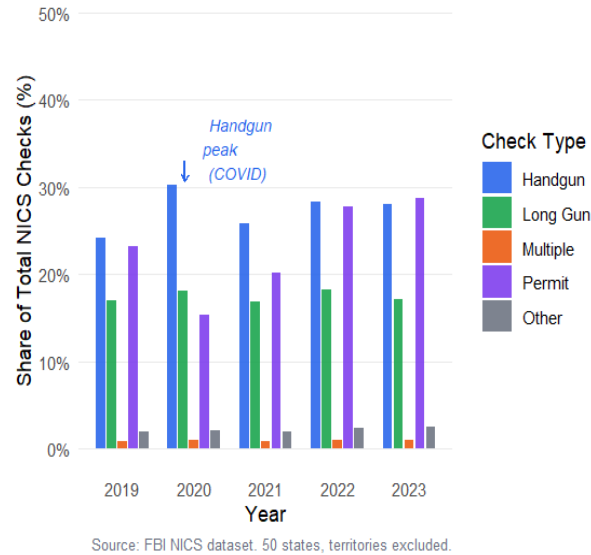


Figure 3. National Composition of Demand Checks

Month	Forecast	Lower 95%	Upper 95%
Oct 2021	2,913,454	2,509,906	3,317,003
Nov 2021	3,237,072	2,717,207	3,756,936
Dec 2021	3,567,060	2,982,604	4,151,516
Jan 2022	3,540,342	2,916,359	4,164,325
Feb 2022	3,194,722	2,545,368	3,844,075
Mar 2022	4,103,528	3,437,370	4,769,686
Apr 2022	3,271,164	2,593,593	3,948,735
May 2022	3,154,411	2,468,910	3,839,912
Jun 2022	3,259,061	2,567,926	3,950,195
Jul 2022	3,067,425	2,372,195	3,762,655
Aug 2022	2,947,578	2,249,300	3,645,855
Sep 2022	2,830,261	2,129,658	3,530,863

Figure 4. 12-Month Forecast